Entra ID ↔ Google Cloud Identity — SSO Federation & Provisioning (GCDS / SCIM)

Implementation Solution Guide

Version 1.0.0 · 2026-07-06

Contents

Executive Summary — Entra ID ↔ Google Cloud Identity: SSO Federation & Provisioning 4

What you get . . . 4

What this delivers . . . 4

The standard deployment model . . . 5

Outcome . . . 5

1. Solution Overview — Entra ID ↔ Google Cloud Identity: SSO Federation & Provisioning 5

Why identity is the foundation of a Google transformation . . . 5

The identity model . . . 5

How the pieces fit together . . . 6

What you can control . . . 6

Benefits . . . 6

Limitations (be honest with buyers) . . . 7

The standard deployment model . . . 7

2. Requirements Guide — Entra ID ↔ Google Cloud Identity SSO & Provisioning 7

2.1 Both control planes accessible (Entra + Google) . . . 8

2.2 Administrative roles (least privilege) . . . 8

2.3 Source of truth + provisioning mechanism decided . . . 8

2.4 Custom domain verified in BOTH Entra and Google . . . 9

2.5 Google admin account with Admin API privileges (for SCIM) . . . 9

2.6 GCDS host (if provisioning from Active Directory) . . . 9

2.7 DNS control for the federated domains . . . 10

2.8 Microsoft Graph PowerShell SDK . . . 10

2.9 A named pilot OU / group and change control . . . 10

2.10 Break-glass accounts (both clouds) . . . 11

Checklist . . . 11

Global Requirements — Google Workspace & Gemini Solutions 11

1. Google Workspace / Cloud Identity tenant . . . 12

2. Source Microsoft 365 / Entra environment (for migration & coexistence) . . . 12

3. DNS control for the routed domains . . . 12

4. Identity & provisioning source of truth . . . 12

5. A pilot scope (OU / group) and change control . . . 13

6. Security, compliance & governance stakeholders . . . 13

3. Implementation Guide — Configure Entra→Google SSO Federation & Provisioning 13

Step 0 — Assess and decide (do this first) . . . 14

Step 1 — Add the gallery app (Entra) . . . 14

Step 2 — Configure SSO federation (Entra IdP → Google, SAML 2.0) . . . 14
Step 3 — Configure provisioning (one source of truth) . . . 15
Step 4 — Pilot: validate sign-in AND provisioning on the named OU/group . . . 17
Step 5 — Expand . . . 17
5. Testing Guide — SSO & Provisioning Conformance Gates 18
Test 1 — Entra-side identity state is inventoried . . . 18
Test 2 — Custom domain matches on both sides . . . 18
Test 3 — SSO federation works (pilot sign-in) . . . 18
Test 4 — Provisioning creates users AND groups . . . 18
Test 5 — One source of truth (no dual mastering) . . . 18
Test 6 — De-provisioning removes access (leaver path) . . . 18
Test 7 — Break-glass path is intact . . . 19
Test 8 — Scope is limited during pilot . . . 19
Test 9 — Coverage across the estate (expansion) . . . 19
Pass/fail gate . . . 19
6. Troubleshooting Guide — Entra↔Google SSO & Provisioning 19
A. Discovery / validation / report returns nothing or errors . . . 20
B. SSO fails — users cannot sign in to Google . . . 20
C. Provisioning — users/groups not appearing in Google (SCIM, Path A) . . . 21
D. Provisioning — GCDS (Path B) . . . 21
E. Two sources of truth — drift and conflicts . . . 21
F. De-provisioning did not remove access . . . 22
G. Coverage report shows gaps . . . 22
H. Diagnostics quick reference . . . 22
I. When to escalate . . . 22
7. Best Practices — Entra↔Google SSO & Provisioning 22
8. Deliverables Summary — Entra↔Google SSO & Provisioning 24
Documentation . . . 24
PowerShell scripts . . . 24
Identity reference doc — outline . . . 25
What the buyer receives . . . 25
Outcome . . . 26
Appendix A — PowerShell Scripts 27

Executive Summary — Entra ID ↔ Google Cloud Identity: SSO Federation & Provisioning

White-label · Google Category

In a Google transformation, **identity is the backbone.** Before mail, files, or apps can move or coexist, users have to sign in **once** and exist **consistently** across both clouds — the same person, the same account, the same groups — with **Microsoft Entra ID** as the enterprise identity provider and **Google Workspace / Cloud Identity** as the relying party. Left unmanaged, identity splits: two logins, accounts present in one directory but not the other, groups that drift, and licensing that no longer matches who is active. Everything downstream inherits that confusion.

This package stands up the identity foundation deliberately and in the right order: **federate Entra ID to Google for single sign-on** (SAML 2.0, via the "Google Cloud / G Suite Connector by Microsoft" gallery app), then **provision users and groups into Google** from a single source of truth — **SCIM push** when Entra masters identity, or **Google Cloud Directory Sync (GCDS)** when on-prem Active Directory does. Accounts, group membership, and licensing stay consistent across Microsoft and Google, and the lifecycle **closes** — leavers are de-provisioned, not orphaned. It is deployed **pilot-first** and **rollback-ready**.

What you get

• A plain-language Solution Overview of the identity model — Entra as IdP federating to Google, the three provisioning options (SCIM push, GCDS pull, JIT-convenience), and the cardinal rule of **one source of truth per object**.

• A Requirements Guide with verify / implement-if-missing for every prerequisite — both control planes, the source-of-truth decision, the custom domain verified in **both** clouds, the SCIM admin account or GCDS host, a named pilot, and break-glass on both sides.

• An Implementation Guide with the exact click-by-click steps to configure **Entra→Google SAML SSO** (both consoles, NameID = user.mail, domain-specific issuer), **provisioning** of users and groups (SCIM Path A or GCDS Path B, including de-provisioning), and to **pilot** sign-in and provisioning on a named OU/group before expanding.

• Five PowerShell scripts for identity-readiness discovery, per-user SSO + provisioning validation, an identity coverage/exception report, safe remediation, and a pilot-scope builder.

• A Testing Guide of conformance gates (sign-in works, users/groups provisioned, one source of truth, de-provisioning removes access, break-glass intact) and a Troubleshooting Guide for a confident, staged rollout.

What this delivers

A stable identity foundation: users authenticate once against Entra ID and reach Google without a second password; users, groups, and licensing stay consistent because one directory is authoritative and provisions the other; and access is removed when a person leaves. SSO is correctly scoped to authentication only, provisioning to a single source of truth, and the whole thing is proven on a pilot before

it touches production — with the Google-side console steps authored from Google's own docs and GCDS internals deferred to Google where appropriate.

The standard deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

One identity across Microsoft and Google — single sign-on and consistent provisioning — underpinning coexistence, migration, and secure access through the transformation and beyond, proven on a named pilot and reversible at every step.

1. Solution Overview — Entra ID ↔ Google Cloud Identity: SSO Federation & Provisioning

Why identity is the foundation of a Google transformation

When an organization adopts Google Workspace / Cloud Identity while Microsoft Entra ID remains the enterprise identity provider, the first thing that has to work is **identity**. Users must sign in **once** and exist **consistently** across both clouds — the same person, the same account, the same groups — or every downstream workload (mail, Drive, Meet, third-party apps) inherits the confusion. The failure mode is a **split identity**: two logins, accounts that exist in one directory but not the other, groups that drift, and licensing that no longer matches who is actually active.

This solution stands up the identity foundation deliberately and in the right order: **federate Entra ID to Google for single sign-on**, then **provision users and groups into Google** from a single source of truth — so accounts, group membership, and licensing stay consistent across Microsoft and Google. It is the base layer that coexistence and migration are built on, deployed **pilot-first** and **rollback-ready**.

> ***Vendor boundary.*** *The **Microsoft / Entra** side (the "Google Cloud / G Suite Connector by Microsoft" gallery application) and the **Google Admin console / Cloud Identity** side are authored at exact click-by-click detail from the vendors' current public documentation. **Google Cloud Directory Sync (GCDS)** is Google's supported tool for AD/LDAP-sourced provisioning; its internal configuration is deferred to Google's own docs where noted. Any capability that is preview or rolling out is flagged "verify in your tenant" — never invented.*

The identity model

Two decisions define the whole design. Make them explicitly, up front.

1. Who authenticates the user (SSO / federation). Entra ID acts as the **identity provider (IdP)**; Google Workspace / Cloud Identity is the **relying party (SP)**. Sign-in is federated over **SAML 2.0** using the Entra gallery app **"Google Cloud / G Suite Connector by Microsoft."** The user authenticates against Entra ID once; Google trusts the SAML assertion. This is **authentication only** — it does not create the Google account and it does not decide Google authorization (OU/group/admin role); those are Google-side.

2. Where the account comes from (provisioning) — one source of truth. An account has to **exist** in Google before (or as) SSO federates a user into it. Choose exactly one authoritative source per object and one mechanism to move it into Google:

- **SCIM push (Entra → Google)** — Entra ID is the source of truth. The same gallery app's

Provisioning capability creates, updates, and de-provisions Google users and groups over SCIM. Choose this when **Entra ID / Microsoft 365** masters identity.

- **GCDS pull (AD/LDAP → Google)** — **Google Cloud Directory Sync** runs on a host with LDAP access

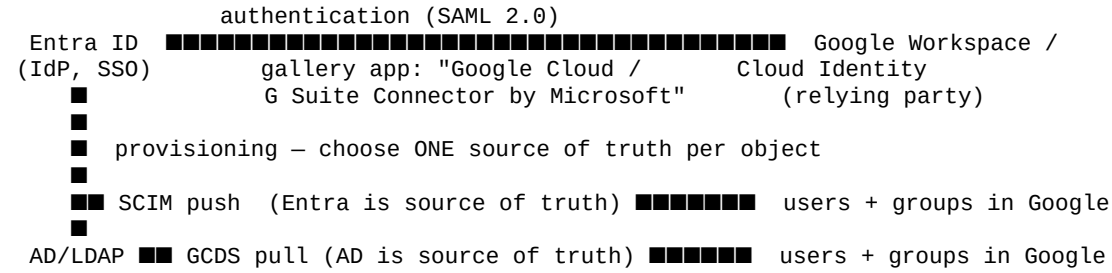
to on-prem Active Directory and syncs users/groups into Google on a schedule. Choose this when **on-prem Active Directory** remains authoritative.

- **JIT (just-in-time)** — SAML JIT can create a Google user on first SSO. Treat JIT as a

convenience, not the source of truth — for lifecycle, and especially **de-provisioning of leavers**, use SCIM push or GCDS so removals actually flow through.

The cardinal rule: **one source of truth per object.** Never let SCIM push (Entra) and GCDS pull (AD) master the same users — they fight and the directories drift.

How the pieces fit together



SSO federates the **sign-in**; provisioning keeps the **accounts and groups** consistent. Both are required — SSO with no provisioning strands users at a login page with no account; provisioning with no SSO leaves two separate passwords.

What you can control

- **Federation** — Entra ID as IdP, Google as relying party, over SAML 2.0, with a NameID that matches

the verified Google domain.

- **Provisioning** — one source of truth (Entra via SCIM, or AD via GCDS), moving users and groups

into Google, with de-provisioning that removes leavers.

- **Consistency** — the same people, groups, and licensing state across Microsoft and Google.
- **Coverage** — per-user visibility into who is federated, who is provisioned, and where the

exceptions are.

Benefits

• **Sign in once** — users authenticate against Entra ID and reach Google without a second password.

• **Consistent accounts and groups** — one source of truth provisions Google; membership and licensing

stay aligned instead of drifting.

• **Lifecycle that closes** — de-provisioning through SCIM/GCDS removes access when a person leaves,

instead of orphaning accounts.

• **A foundation to build on** — coexistence and migration inherit stable identity instead of fixing

it under pressure later.

• **Pilot-first and reversible** — federation and provisioning are proven on a named pilot OU/group

before org-wide enforcement, with a documented rollback.

Limitations (be honest with buyers)

• SSO is **authentication only.** It does not create Google accounts (provisioning does) and does not

grant Google admin — Google decides authorization by OU/group/admin role.

• **One source of truth is mandatory.** SCIM push and GCDS pull must not master the same objects.

JIT is not a lifecycle mechanism — leavers must be de-provisioned by SCIM or GCDS.

• The NameID / unique identifier in the SAML assertion **must match the verified Google domain and the

Google account's email**, or users cannot sign in.

• The scripts inventory and validate the **Microsoft / Entra side** through Microsoft Graph

(read-only). **Google-side** state — the SSO profile, provisioned users/groups, GCDS runs — is confirmed in the Google Admin console and GCDS.

• Some Entra provisioning, conditional-access, and PIM-for-groups capabilities require **Entra ID

P1/P2** licensing (see Requirements).

The standard deployment model

> *Assess the identity estate and pick the source of truth → configure Entra→Google SSO federation and provisioning → pilot both on a named OU/group → validate sign-in and provisioning end-to-end → expand across the organization, tracking exceptions. Repeatable pattern:* ***Assess*** *→* ***Configure*** *→* ***Pilot*** *→* ***Validate*** *→* ***Expand****, with explicit pass/fail gates and a documented rollback for every production change.*

2. Requirements Guide — Entra ID ↔ Google Cloud Identity SSO & Provisioning

Verify every requirement before configuring federation or provisioning. Each entry is **Description / Why required / How to verify / How to implement if missing.** Category-wide prerequisites (Google tenant, source M365/Entra environment, DNS control, source of truth, pilot scope, governance stakeholders) are in `global-requirements-google.md` — this guide adds what the Entra↔Google SSO + provisioning work needs on top of those.

2.1 Both control planes accessible (Entra + Google)

- **Description:** Administrative access to **Microsoft Entra ID** at **entra.microsoft.com** (to add

and configure the "Google Cloud / G Suite Connector by Microsoft" gallery app) and to the **Google Admin console** at **admin.google.com** as a **super-admin** (to accept the SAML SSO profile and to authorize SCIM provisioning).

- **Why required:** Federation and provisioning are configured on **both** ends — Entra publishes the

SAML IdP details and runs the provisioning service; Google accepts the SSO profile and authorizes the provisioning connection.

- **How to verify:** Sign in to **entra.microsoft.com** with an app-management role, and to

admin.google.com as a super-admin.

- **How to implement if missing:** Obtain the roles from the respective tenant owners before starting.

2.2 Administrative roles (least privilege)

- **Description:** On **Entra**, one of `Application Administrator`, `Cloud Application Administrator`,

or the app's `Application Owner` to configure SSO and provisioning; reserve `Global Administrator` for one-time tenant settings only. On **Google**, a **super-admin** to configure the SSO profile, plus a dedicated Google admin account (see 2.5) for the SCIM connection.

- **Why required:** Configuring federation and provisioning are privileged actions on each plane;

least-privilege roles limit blast radius.

- **How to verify:** Entra → **Roles and administrators**; Google → **Account** → **Admin roles**.

- **How to implement if missing:** Assign the least-privilege role on each plane before configuring.

2.3 Source of truth + provisioning mechanism decided

- **Description:** A documented decision on **which directory masters users and groups** and **which

mechanism **moves them into Google:** SCIM push **(Entra ID is source of truth),** GCDS pull **(on-prem Active Directory is source of truth), or** JIT** on first SSO (convenience only, not lifecycle). One authoritative source per object — never two.

- **Why required:** Two sources of truth is the anti-pattern that makes directories drift and conflict;

the mechanism you pick determines the entire provisioning configuration and the de-provisioning path.

- **How to verify:** A one-line statement exists: "Google users/groups are mastered by ___ and

provisioned via ___." (Enterprise-source-of-truth detail is in global requirement 4.)

• **How to implement if missing:** Decide the source of truth and the SCIM / GCDS / JIT mechanism per object type before enabling any provisioning.

2.4 Custom domain verified in BOTH Entra and Google

• **Description:** Every user to be provisioned/federated must have a **UPN on a verified custom domain **(e.g.** `user@contoso.com`**,** not `user@contoso.onmicrosoft.com`**), and that domain must be** verified in Google **as well (Google Admin console** → Account → Domains → Manage domains**).

• **Why required:** Google **rejects** `*.onmicrosoft.com` usernames for provisioning, and SSO fails if the SAML NameID does not match a verified Google domain. The user's identifier must match on both sides.

• **How to verify:** In Entra, users' UPNs are on the custom domain; in Google, the domain shows as verified under **Manage domains**.

• **How to implement if missing:** Add and verify the custom domain in Entra, change affected UPNs to it, then add and verify the same domain in Google (**Add a domain** → **ADD DOMAIN & START VERIFICATION**).

2.5 Google admin account with Admin API privileges (for SCIM)

• **Description:** If provisioning via **SCIM push**, a Google admin account whose admin role has **all Admin API privileges **enabled — this is the account Entra authorizes for the provisioning connection. Also enable** Security → Access and data control → API Controls → "Trust internal, domain-owned apps."**

• **Why required:** Entra's provisioning service authorizes against this account to create/update/remove Google users and groups over the Directory API; without the Admin API privileges the connection test fails.

• **How to verify:** Google Admin console → **Account** → **Admin roles** → the role's **Privileges** show Admin API privileges enabled; API Controls has the trust setting saved.

• **How to implement if missing:** Edit the role's privileges to enable Admin API privileges; save the API Controls trust setting. (Not required for GCDS-only or JIT-only designs.)

2.6 GCDS host (if provisioning from Active Directory)

• **Description:** If provisioning via **GCDS pull**, a **Windows or Linux host with LDAP access to Active Directory **to run** Google Cloud Directory Sync **— best on a** separate machine**, not the domain controller — plus a Google admin identity GCDS authenticates with.

• **Why required:** GCDS reads AD/LDAP and writes users/groups into Google on a schedule; it needs network line-of-sight to AD and a place to run its scheduled sync.

• **How to verify:** The host can reach AD over LDAP(S) and outbound to Google; GCDS is installed (or a

target host is nominated).

• **How to implement if missing:** Nominate/provision the host, grant it LDAP access to AD, install GCDS, and prepare a Google admin for it. (Not required for SCIM-only or JIT-only designs.)

2.7 DNS control for the federated domains

• **Description:** Ability to create/modify **DNS records** (TXT/CNAME for domain verification) for the domains being federated and provisioned. (Category-wide DNS control is global requirement 3.)

• **Why required:** Verifying the custom domain in Google (2.4) requires a DNS change; identity changes are staged under change control.

• **How to verify:** Confirm access to the authoritative DNS zone for each domain.

• **How to implement if missing:** Coordinate with whoever controls DNS; stage under change control.

2.8 Microsoft Graph PowerShell SDK

• **Description:** The Microsoft Graph PowerShell SDK on the operator workstation, with consent to the read scopes the scripts use.

• **Why required:** Discovery, validation, and reporting read the Entra-side identity/SSO/provisioning state — the gallery app's service principal, its SSO mode, its SCIM synchronization job, assignments, and users — through Microsoft Graph.

• **How to verify:** `Get-Module -ListAvailable Microsoft.Graph.Authentication` returns a version; a test `Connect-MgGraph` succeeds.

• **How to implement if missing:** `Install-Module Microsoft.Graph -Scope CurrentUser`. The read-only scripts request least privilege (`Application.Read.All`, `Directory.Read.All`, `Policy.Read.All`, `Synchronization.Read.All`, `User.Read.All`); the scope helper adds `Group.ReadWrite.All` only with `-Create`; remediation adds `Synchronization.ReadWrite.All` / `AppRoleAssignment.ReadWrite.All`.

2.9 A named pilot OU / group and change control

• **Description:** A **named pilot organizational unit or group** (a handful of representative users) to validate SSO and provisioning on before org-wide enforcement, and a **change-control** window for the production identity changes.

• **Why required:** Federation and provisioning are proven on a pilot first (Assess → Configure → Pilot → Validate → Expand); enforcing SSO org-wide before validating can lock users out.

• **How to verify:** A named pilot OU/group exists and a change window/approval path is available.

• **How to implement if missing:** Create the pilot OU/group; use `GWSSO-Set-ScopeTargeting.ps1` to

build the Entra pilot-scope group, and scope the Google SSO profile to the matching Google OU/group.

2.10 Break-glass accounts (both clouds)

• **Description:** On **Google**, a **cloud-only super-admin excluded from the SSO profile** so an SSO change can never lock admins out. On **Entra**, two cloud-only `@onmicrosoft.com` emergency-access admins excluded from broad conditional-access policies.

• **Why required:** A federation change could interfere with sign-in on either side; break-glass paths restore access.

• **How to verify:** Google → a super-admin with the SSO profile set to **none**; Entra → two cloud-only admins excluded from broad policies.

• **How to implement if missing:** Create them, exclude them from SSO/CA policies, store credentials securely, and monitor their sign-ins.

Checklist

- Both control planes accessible — Entra (app-mgmt role) + Google (super-admin)
- Least-privilege admin roles assigned on each plane
- Source of truth + provisioning mechanism decided (SCIM push / GCDS pull / JIT) per object
- Custom domain verified in **both** Entra and Google; no `*.onmicrosoft.com` UPNs in scope
- Google admin account with Admin API privileges + API Controls trust set (if SCIM)
- GCDS host with LDAP access to AD prepared (if GCDS)
- DNS control confirmed for the federated domains
- Microsoft Graph PowerShell SDK installed and consented
- Named pilot OU/group + change window — `GWSSO-Set-ScopeTargeting.ps1`
- Break-glass: Google super-admin excluded from SSO + two Entra cloud-only admins

Global Requirements — Google Workspace & Gemini Solutions

These requirements apply to **every** solution in the **Google** category. They are the common foundation each Google Workspace / Gemini solution assumes is already in place (or stands up as part of its own Requirements section). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Google solution).*** *The **Google Admin console** (admin.google.com), **Google Cloud / Cloud Identity**, and the **Microsoft / Entra** sides are authored at exact, current click-by-click detail from the vendors' public documentation.* ***Gemini Enterprise***

*capabilities that are in preview or rolling out are **cited or flagged "verify in your tenant"** — never invented. **Third-party migration tools** (e.g. for mail/Drive/Chat data movement) and **third-party enterprise apps** (ServiceNow, SAP, Databricks, etc.) are integrated at the pattern/architectural level; their internal console steps are deferred to the vendor's own docs.*

1. Google Workspace / Cloud Identity tenant

• **Description:** A provisioned Google Workspace (or Cloud Identity) tenant with a **super-admin** account for configuration, and the appropriate Workspace edition/licensing for the capability in scope (e.g. Enterprise Plus for Vault/CAA/DLP, Gemini/AI add-on where required).

• **Why required:** All configuration, policy, and API work is performed against this tenant.

• **How to verify:** Sign in to **admin.google.com** as a super-admin; confirm edition and license counts under **Billing / Subscriptions**.

• **How to implement if missing:** Provision the tenant and licenses through Google or an authorized reseller before starting.

2. Source Microsoft 365 / Entra environment (for migration & coexistence)

• **Description:** For migration/coexistence/identity solutions, administrative access to the source **Microsoft 365** tenant and **Microsoft Entra ID** (Global Administrator or the least-privilege roles the solution names).

• **Why required:** Migration, directory sync, SSO federation, and coexistence read from and change the Microsoft side.

• **How to verify:** Confirm access to the M365 admin center and the Entra admin center with the required role.

• **How to implement if missing:** Obtain the role from the M365/Entra tenant owner.

3. DNS control for the routed domains

• **Description:** Ability to create/modify **DNS records** (MX, SPF, DKIM, DMARC, TXT, CNAME) for the domains being migrated, coexisting, or verified.

• **Why required:** Domain verification, mail routing/coexistence, and authentication all depend on DNS changes.

• **How to verify:** Confirm access to the authoritative DNS zone for each domain.

• **How to implement if missing:** Coordinate with whoever controls DNS; stage changes under change control (mail-flow changes are high-impact).

4. Identity & provisioning source of truth

• **Description:** A defined **identity source of truth** (on-prem AD, Entra ID, or an HR/IGA system)

and the sync mechanism (e.g. **Google Cloud Directory Sync (GCDS)** or SCIM-based provisioning).

- **Why required:** Consistent users/groups across ecosystems underpin SSO, licensing, and access.
- **How to verify:** Confirm which directory is authoritative and how objects flow into Google.
- **How to implement if missing:** Define the source of truth and provisioning model before configuring SSO or migrating.

5. A pilot scope (OU / group) and change control

- **Description:** A **pilot organizational unit or group** to validate on before fleet-wide rollout, and a **change-control** process for production changes (especially mail flow, DNS, identity, and DLP).
- **Why required:** Every solution follows an **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model; the pilot proves the design and change control protects production.
- **How to verify:** Confirm a named pilot OU/group exists and a change window/approval path is available.
- **How to implement if missing:** Create a pilot OU/group and agree the change-control process with the owners before making production changes.

6. Security, compliance & governance stakeholders

- **Description:** Engagement with the **security (CISO), legal/eDiscovery, and data-governance** stakeholders whose policies the configuration must satisfy (DLP, retention, Vault/Purview, and any regulatory regime such as ITAR/DFARS/Assured Controls where applicable).
- **Why required:** Data governance, eDiscovery, DLP, and regulated-workload settings must align to enterprise policy and legal requirements, not just technical defaults.
- **How to verify:** Confirm the applicable policies and named stakeholders/approvers.
- **How to implement if missing:** Identify the owners and obtain the policy/compliance requirements before configuring governance controls.

*The deployment model across every Google solution is the same: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes.*

3. Implementation Guide — Configure Entra→Google SSO Federation & Provisioning

This guide configures the identity foundation in the right order: **assess and pick the source of truth** → **configure SSO federation (Entra IdP → Google)** → **configure provisioning (SCIM push or GCDS pull)** → **pilot both on a named OU/group** → **validate sign-in and provisioning** → **expand.** Every change is staged, validated on the pilot, and reversible.

> ***Vendor boundary.*** *The **Entra** side is authored from Microsoft's current public docs (the "Google Cloud / G Suite Connector by Microsoft" gallery app); the **Google Admin console / Cloud Identity** side from Google's public docs. **GCDS** internals are Google's tool; its detailed field-by-field configuration is deferred to Google's own documentation where noted. Menu names and URLs change — treat any exact label as "verify in your current console."*
>
> ***Order matters.*** *Provision (or verify accounts already exist) **before** enforcing SSO org-wide — SSO federates a sign-in into an account that must already exist. Always keep a break-glass Google super-admin **outside** the SSO profile (Requirement 2.10).*

Step 0 — Assess and decide (do this first)

1. Run `GWSSO-Discovery.ps1` to inventory the **Entra side**: whether the "Google Cloud / G Suite

Connector by Microsoft" app exists, its SSO mode (SAML / none), whether a SCIM provisioning (synchronization) job is present, how many users/groups are assigned, and the tenant's verified custom domains.

1. Confirm the **source of truth** (Requirement 2.3): **Entra** → **SCIM push**, **AD** → **GCDS pull**, or

JIT (convenience only). Write it down; the rest of the guide branches on it.

1. Confirm the **custom domain is verified in both Entra and Google** and that no in-scope user has a

`*.onmicrosoft.com` UPN (Requirement 2.4).

1. Build the pilot scope with `GWSSO-Set-ScopeTargeting.ps1` and identify the matching Google pilot

OU/group.

Step 1 — Add the gallery app (Entra)

1. Sign in to **entra.microsoft.com** as at least a **Cloud Application Administrator**.
2. Browse to **Entra ID** → **Enterprise apps** → **New application**.
3. In **Add from the gallery**, search **"Google Cloud / G Suite Connector by Microsoft"**, select it,

and add it. (If you already use this app for SSO, you can reuse it for provisioning; when first testing, a separate app instance is fine.)

Step 2 — Configure SSO federation (Entra IdP → Google, SAML 2.0)

2a. Entra side — SAML SSO

1. In the app, open **Single sign-on** and select **SAML**.
2. Edit **Basic SAML Configuration**. Use one of the documented patterns (verify against your console):

• **Identifier (Entity ID):** `google.com` (no domain-specific issuer) **or**

`google.com/a/your-domain-com` (domain-specific issuer). This must **match** the domain-specific issuer choice you make on the Google side in 2b.

- **Reply URL (ACS):** for Workspace/Gmail `https://www.google.com`; for Google Cloud Platform

`https://www.google.com/acs` (or the `/a/your-domain-com` variants).

- **Sign on URL:**

`https://www.google.com/a/your-domain-com/ServiceLogin?continue=https://mail.google.com`.

1. **Attribute / NameID:** the **Unique User Identifier (NameID)** must be the user's **email address**

that matches their Google account — map it to **user.mail** (not the default `user.userprincipalname`). This is the single most common cause of "cannot sign in."

1. In **SAML Signing Certificate**, download **Certificate (Base64)**.
2. In **Set up ...**, copy the **Login URL** and **Logout URL** for the Google side. (Note: Microsoft

documents the correct sign-out URL as `https://login.microsoftonline.com/common/wsfederation?wa=wsignout1.0`.)

2b. Google side — accept the SAML SSO profile

1. Sign in to **admin.google.com** as a **super-admin**.
2. Go to **Menu** → **Security** → **Authentication** → **SSO with third-party IdP**.
3. On the **Third-party SSO profile for your organization** tab:

- **Sign-in page URL** ← Entra **Login URL**.
- **Sign-out page URL** ← Entra **Logout URL**.
- **Verification certificate** ← upload the Base64 certificate from 2a.
- **Use a domain-specific issuer** ← check/uncheck to match your Entra Identifier choice in 2a.
- **Change password URL** ← `https://mysignins.microsoft.com/security-info/password/change`.
- **Save.**

1. **Scope the SSO profile to the pilot first.** Google supports assigning the SSO profile per **user,

OU, or group**. Assign the profile to the** pilot OU/group **and set it to** none** for everyone else (and for the break-glass super-admin) so only the pilot federates until you validate.

> ***What SSO does and does not do.*** *The SAML assertion* ***authenticates****; it does not create the Google account and does not grant Google admin. The account must already exist (Step 3), and Google decides authorization by OU/group/admin role.*

Step 3 — Configure provisioning (one source of truth)

Do **one** of the following, per the decision in Step 0.

Path A — SCIM push (Entra ID is the source of truth)

1. **Google prerequisites** (Requirement 2.5): **Security → Access and data control → API Controls → "Trust internal, domain-owned apps" → SAVE**; and a Google admin account whose** Admin role **(**Account → Admin roles**) has** all Admin API privileges** enabled.

1. In Entra, open the app → **Provisioning** tab → **Get started** → **+ New configuration**.

2. Under **Admin Credentials**, select **Authorize** → in the Google dialog, **Accept** to grant Entra permission to manage the Google tenant → **Test Connection** → **Create**.

1. On **Overview** → **Properties**, enable **notification emails**, set a **quarantine** address, and enable **accidental-deletion prevention**; **Apply**.

1. **Attribute Mapping** → **users:** review mappings; **primaryEmail** is the matching attribute (and must equal the SSO NameID). **Attribute Mapping** → **groups:** review group mappings (email, name, description, members) so **groups and memberships** provision too.

1. **Scope who provisions:** either **assign** users/groups to the app (Enterprise app → Users and groups) or set an **attribute-based scoping filter**. **Start with the pilot OU/group only.**

1. Use **on-demand provisioning** to validate a couple of pilot users, then **Start provisioning** from **Overview**.

• **De-provisioning:** removing a user from scope (or disabling in Entra) **suspends** them in Google — it does **not** hard-delete the Google object. That is the intended leaver path.

Path B — GCDS pull (on-prem Active Directory is the source of truth)

1. Install **Google Cloud Directory Sync** on the prepared host with LDAP access to AD (Requirement 2.6) — best on a **separate machine**, not the domain controller.

1. In **GCDS Configuration Manager**, connect to **Active Directory** (LDAP) and to **Google (Cloud Identity / Workspace)** with a Google admin.

1. Under **General Settings**, select **User Accounts** (and **Groups** if syncing groups). Configure **User Accounts** → **Additional User Attributes** and the OU/attribute mapping rules that decide which objects flow. (Field-by-field mapping is per Google's GCDS docs.)

1. **Simulate** a sync — GCDS reports the changes it *would* make without applying them; review the simulation against expectations.

1. Run a real sync for the **pilot OU** only, then **schedule** GCDS for continuous provisioning.

• **De-provisioning:** removing/disabling the AD object flows through GCDS to suspend the Google account on the next run.

JIT note

SAML **just-in-time** creation makes a Google user on first SSO. It is a **convenience, not the source of truth** — for lifecycle, and especially **de-provisioning leavers**, use Path A or Path B so removals actually propagate. Do not rely on JIT alone.

> ***Never run both Path A and Path B against the same objects.*** *SCIM push (Entra) and GCDS pull (AD) mastering the same users is two sources of truth — they conflict and drift. Pick one per object.*

Step 4 — Pilot: validate sign-in AND provisioning on the named OU/group

1. Confirm the **pilot users exist in Google** (provisioned via Path A/B, or created for the test).

2. A pilot user **signs in** via Entra SSO (use **Test this application** in Entra, or the Google

sign-on URL). Expected: redirect to Entra, authenticate, land in Google — no second password.

1. **Provisioning check:** add a pilot user to a source group; confirm the user **and group membership**

appear in Google. Remove the user; confirm they are **de-provisioned (suspended)** in Google.

1. Run `GWSSO-Validation.ps1` to confirm the Entra side: SSO mode is SAML, a provisioning job exists

and is running, and the pilot users are assigned/in scope.

1. Confirm the **break-glass Google super-admin** (SSO profile = none) can still sign in outside SSO.

Gate: do not proceed to expand until sign-in works, provisioning creates and de-provisions correctly, and break-glass is confirmed (see the Testing Guide gates).

Step 5 — Expand

1. Widen the **Google SSO profile** assignment from the pilot OU/group to the next batches of OUs/groups

(leaving break-glass at **none**).

1. Widen **provisioning scope** (SCIM assignment/filter, or GCDS OU rules) in the same batches.

2. Re-run `GWSSO-Discovery.ps1`, `GWSSO-Validation.ps1`, and `GWSSO-Report.ps1` on a cadence so

coverage and exceptions stay visible as the rollout grows.

1. Record every deliberate exception (e.g. service accounts intentionally not federated) with an owner.

Order of operations

`Assess + pick source of truth → add gallery app → configure SAML SSO (Entra + Google) → configure provisioning (SCIM push OR GCDS pull) → pilot (sign-in + provisioning on named OU/group) → validate → expand in batches, tracking exceptions`

Rollback

• **SSO:** set the Google SSO profile to **none** for the affected OU/group (or turn the Entra SSO app

off) — sign-in reverts to Google-native or the prior IdP. The break-glass super-admin is always outside SSO.

• **Provisioning (SCIM): Stop provisioning** in the Entra app — existing Google objects remain; no new writes. Re-scoping to fewer users is non-destructive.

• **Provisioning (GCDS):** disable the scheduled task / narrow the OU rules — existing objects remain.

• Every change is piloted on a named OU/group and reversible before org-wide enforcement.

5. Testing Guide — SSO & Provisioning Conformance Gates

Run after the pilot (Implementation Step 4) and on the recurring cadence during and after expansion. Each test checks whether the **identity foundation** works the standard way — not just that one user can sign in, but that federation, provisioning, and de-provisioning are all correct and that break-glass is intact.

Test 1 — Entra-side identity state is inventoried

Do: Run `GWSSO-Discovery.ps1`. **Expected:** The "Google Cloud / G Suite Connector by Microsoft" app is listed with its SSO mode (SAML), whether a SCIM provisioning job exists, the count of assigned users/groups, and the tenant's verified custom domains — a usable baseline. No changes made.

Test 2 — Custom domain matches on both sides

Do: Confirm the in-scope custom domain is verified in **Entra** and in **Google** (Admin console → **Account** → **Domains** → **Manage domains**); confirm no in-scope user has a `*.onmicrosoft.com` UPN. **Expected:** The domain is verified in both clouds; every in-scope user's UPN is on the custom domain. A `*.onmicrosoft.com` UPN is a fail — Google rejects it for provisioning and SSO.

Test 3 — SSO federation works (pilot sign-in)

Do: A **pilot-OU** user signs in via Entra SSO (Entra **Test this application**, or the Google sign-on URL). Run `GWSSO-Validation.ps1` for the Entra-side signal. **Expected:** The user is redirected to Entra, authenticates, and lands in Google with **no second password**. Entra SSO mode is **SAML**; the **NameID = user.mail** and matches the Google domain.

Test 4 — Provisioning creates users AND groups

Do: Add a test user to a source group in scope; run `GWSSO-Validation.ps1` (provisioning section); confirm in the Google Admin console. **Expected:** The user **and the group + membership** appear in Google via the chosen mechanism (SCIM push or GCDS pull). `primaryEmail` matches the SSO NameID. Attributes populate per the mapping.

Test 5 — One source of truth (no dual mastering)

Do: Run `GWSSO-Validation.ps1` and confirm the design. **Expected:** Each object type has exactly **one** master — SCIM push (Entra) **or** GCDS pull (AD), never both for the same users. No object is written by two mechanisms. JIT is not being used as the lifecycle source.

Test 6 — De-provisioning removes access (leaver path)

Do: Remove the test user from the source group (or disable them in Entra/AD); wait for the next cycle/run. **Expected:** The user is **de-provisioned (suspended)** in Google — access is removed, not orphaned. (SCIM removal from scope suspends rather than hard-deletes; GCDS suspends on the next run.) This is the test that proves lifecycle actually closes.

Test 7 — Break-glass path is intact

Do: Sign in as the cloud-only **Google super-admin** whose SSO profile is set to **none**; confirm the two **Entra** cloud-only emergency admins are excluded from broad CA policies. **Expected:** The Google super-admin signs in **outside** SSO; an SSO or provisioning change cannot lock admins out.

Test 8 — Scope is limited during pilot

Do: Confirm the **Google SSO profile** is assigned only to the pilot OU/group (everyone else and break-glass = **none**) and provisioning scope (SCIM assignment/filter or GCDS OU rules) covers only the pilot. **Expected:** Only the pilot federates and provisions until the gate passes — the rest of the org is untouched.

Test 9 — Coverage across the estate (expansion)

Do: Run `GWSSO-Report.ps1`. **Expected:** A per-user/coverage view — federated vs not, provisioned vs not — with the exception list containing only **deliberate, documented** exceptions (e.g. service accounts intentionally not federated). No silent gaps.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–8 pass on the pilot OU/group — sign-in works (Test 3), provisioning creates users+groups (Test 4), one source of truth (Test 5), de-provisioning removes access (Test 6), break-glass intact (Test 7), scope limited to pilot (Test 8)
Expand pass	Tests 1–7 and 9 pass across each batch; the exception list is reviewed and every item is deliberate; discovery / validation / report run on a schedule

6. Troubleshooting Guide — Entra↔Google SSO & Provisioning

Use `GWSSO-Discovery.ps1` for the Entra-side identity state, `GWSSO-Report.ps1` for coverage and exceptions, `GWSSO-Validation.ps1` for per-user SSO/provisioning conformance, and `GWSSO-Remediation.ps1` (preview first) for safe fixes. **Google-side** state — the SSO profile, provisioned users/groups, GCDS runs — is confirmed in the Google Admin console and GCDS.

A. Discovery / validation / report returns nothing or errors

Symptom	Cause	Remediation
`Connect-MgGraph` fails or scopes missing	SDK not installed, or read scopes not consented	`Install-Module Microsoft.Graph -Scope CurrentUser`; re-run and accept `Application.Read.All`, `Directory.Read.All`, `Policy.Read.All`, `Synchronization.Read.All`, `User.Read.All`
The Google gallery app is not found	The "Google Cloud / G Suite Connector by Microsoft" app is not added, or you are in the wrong tenant	Add it from the gallery (Implementation Step 1); confirm the tenant
Provisioning job not listed	`Synchronization.Read.All` not granted, or provisioning never configured	Re-run and consent the synchronization read scope; if truly none, configure provisioning (Path A)

B. SSO fails — users cannot sign in to Google

Symptom	Cause	Remediation
"cannot sign in" / wrong user after enabling SSO	**NameID does not match** the Google account email / verified domain	In the Entra app, map the **Unique User Identifier (NameID)** to **user.mail** (not user.userprincipalname); ensure it equals the Google `primaryEmail` on a verified domain
Users on `*.onmicrosoft.com` fail	Google rejects `*.onmicrosoft.com` identities	Move affected UPNs to the verified custom domain; re-test
SSO loop or "invalid assertion"	Identifier/Entity ID vs domain-specific-issuer mismatch, or ACS/certificate wrong	Make the Entra **Identifier** and the Google **"Use a domain-specific issuer"** choice agree; re-copy the ACS/Reply URL and re-upload the Base64 certificate
Wrong sign-out behavior	Default logout URL is incorrect	Use `https://login.microsoftonline.com/common/wsfederation?wa=wsignout1.0` as the sign-out URL
Admin locked out of Google	SSO enforced with no break-glass path	Sign in as the cloud-only Google **super-admin** whose SSO profile is **none**; keep this path always available

C. Provisioning — users/groups not appearing in Google (SCIM, Path A)

Symptom	Cause	Remediation
Test Connection / Authorize fails	Google admin account lacks **Admin API privileges**, or API Controls trust not set	Google **Account** → **Admin roles** → enable all **Admin API privileges**; **Security** → **Access and data control** → **API Controls** → **"Trust internal, domain-owned apps"** → **SAVE**; re-**Authorize**
Users not provisioned	Users not **assigned**/in scope, or provisioning not started	Assign the users/groups to the app (or set a scoping filter); **Start provisioning**; use **on-demand provisioning** to test a user
Groups exist but memberships/attributes wrong	Group or user attribute mapping incomplete	Review **Attribute Mapping** → **users** (matching = `primaryEmail`) and → **groups** (email, name, members)
Provisioning went to quarantine	Repeated failures (often auth/permission)	Check the **provisioning logs** and quarantine email; fix the Google admin permission/credential and resume
A user has an existing consumer Google account on that email	Conflicting personal account	Use Google's **Transfer Tool** for unmanaged accounts before syncing that user

D. Provisioning — GCDS (Path B)

Symptom	Cause	Remediation
Simulation shows unexpected deletes/changes	OU/attribute mapping rules too broad or misconfigured	Correct the GCDS mapping rules; re-**simulate** and review before any real sync (per Google's GCDS docs)
GCDS cannot read AD	No LDAP line-of-sight / wrong bind account	Run GCDS on a host with LDAP access to AD (not the DC); fix the LDAP connection settings
Sync not recurring	No scheduled task	Schedule GCDS for continuous provisioning after the pilot sync validates

E. Two sources of truth — drift and conflicts

Symptom	Cause	Remediation
Objects flip-flop / conflict	Same users mastered by **both** SCIM push (Entra) and GCDS pull (AD)	Pick **one** master per object; disable the other mechanism for those objects (Requirement 2.3)
Leavers keep access	JIT used as the "source of truth"; no de-provisioning	Use SCIM push or GCDS for lifecycle so removals flow through; treat JIT as convenience only

F. De-provisioning did not remove access

Symptom	Cause	Remediation
Removed user still active in Google	Expected behavior — SCIM **suspends**, does not hard-delete; or the user was removed from *scope* without disabling	Confirm the user is **suspended** in Google (access removed). Removing from sync scope disables, it does not delete the object; delete in Google only per your retention policy
GCDS did not suspend the leaver	AD object not removed/disabled, or next run not executed	Disable/remove the AD object; run/await the next GCDS sync

G. Coverage report shows gaps

- A user who is **federated but not provisioned** (or vice versa) is a gap. Decide: bring them into scope with `GWSSO-Remediation.ps1` (preview, confirm, apply), or record a **deliberate exception** (e.g. a service account intentionally not federated) with an owner and review date. The report should never contain silent, unexplained gaps.

H. Diagnostics quick reference

```
Connect-MgGraph -Scopes 'Application.Read.All','Directory.Read.All','Policy.Read.All','Synchronization.Rea
# find the Google gallery app service principal + its SSO mode
Get-MgServicePrincipal -Filter "startswith(displayName,'Google')" | Select-Object DisplayName, AppId, Pref
# is there a SCIM provisioning (synchronization) job on it?
Get-MgServicePrincipalSynchronizationJob -ServicePrincipalId your-service-principal-id
# verified custom domains (must match Google; no onmicrosoft.com for in-scope users)
Get-MgDomain | Where-Object IsVerified | Select-Object Id, IsDefault
```

I. When to escalate

If admins are locked out of Google and the break-glass super-admin path also fails, or provisioning is mass-suspending users unexpectedly, **stop the change**: set the SSO profile to **none** for the affected scope and **Stop provisioning** in the Entra app (existing objects remain), then engage the Google Workspace admin owner and your change-control/incident process before re-enabling.

7. Best Practices — Entra↔Google SSO & Provisioning

- **Decide the source of truth first — one master per object.** Before touching a console, state which directory masters users and groups (Entra via SCIM, or AD via GCDS) and provision outward from it. **Never** let SCIM push and GCDS pull master the same objects — two sources of truth is the drift-and-conflict anti-pattern.
- **Provision before you enforce SSO.** SSO federates a sign-in into an account that must already exist. Provision (or verify accounts) first, then enforce federation — otherwise users authenticate successfully and land nowhere.
- **Match the identifier on both sides.** The SAML **NameID must be the user's email (user.mail)** and must equal the Google account's `primaryEmail` on a **verified custom domain**. A mismatched NameID — or a `*.onmicrosoft.com` UPN — is the number-one cause of "cannot sign in." Verify the domain in both Entra and Google.
- **Keep a break-glass Google super-admin outside SSO — always.** Set that account's SSO profile to **none** so a federation change can never lock admins out. Keep two cloud-only Entra emergency admins excluded from broad conditional-access policies too.
- **Pilot on a named OU/group, then expand in batches.** Assign the Google SSO profile and the provisioning scope to the pilot only; set everyone else to **none** until the gate passes. Expand OU by OU, not all at once.
- **Test the leaver path, not just the joiner path.** Provisioning that creates accounts but never removes them is worse than none. Prove that removing a user from scope (or disabling them) actually **de-provisions/suspends** them in Google — every cycle.
- **Treat JIT as convenience, never as lifecycle.** Just-in-time SSO creation is handy, but it does not de-provision leavers. Use SCIM push or GCDS for the real lifecycle.
- **Drive Google authorization from groups, not per-user admin.** Provision groups and memberships, and let Google OU/group membership decide access and admin roles — so entitlements follow directory membership and reviews, not hand-set per-user grants. (For privileged Google groups, consider Entra **PIM for groups** for just-in-time membership.)
- **Scope provisioning tightly and watch the quarantine channel.** Use assignment or an attribute scoping filter so only intended users flow; enable Entra provisioning **notification emails**, **accidental-deletion prevention**, and a **quarantine** address, and act on quarantine promptly.
- **Run GCDS off the domain controller and simulate before you sync.** Put GCDS on a separate host with LDAP access to AD; always run a **simulation** and review the proposed changes before a real sync.
- **Stage every identity change under change control.** Federation and provisioning changes are high-impact and can lock out or orphan users. Change-window them, keep the rollback ready (SSO profile → none; **Stop provisioning**), and confirm break-glass before each change.
- **Make coverage recurring, not one-time.** Re-run discovery, validation, and the coverage/exception

report on a cadence so new users, new OUs, and any drift are caught and classified as they appear.

- **Maintain the identity reference doc.** Keep the source-of-truth statement, the domain/NameID mapping, the SSO-profile assignment plan, the provisioning scope, and the exception register current — it is the identity foundation's source of truth for coexistence and migration to build on.

8. Deliverables Summary — Entra↔Google SSO & Provisioning

Documentation

Document	Purpose
Executive Summary	Business case and what is in the box
1. Solution Overview	The identity model: Entra ID as IdP federating to Google (SAML SSO), provisioning via SCIM push or GCDS pull, one source of truth, and how the pieces fit
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing (both control planes, source of truth, domain verified in both clouds, SCIM admin / GCDS host, pilot, break-glass)
Global Requirements (Google)	Category-wide prerequisites, published once
3. Implementation Guide	Configure Entra→Google SAML SSO (both consoles), provisioning of users+groups via SCIM push (Path A) or GCDS pull (Path B), pilot sign-in + provisioning on a named OU/group, then expand
5. Testing Guide	SSO + provisioning conformance gates with pass/fail (sign-in, create, one source of truth, de-provisioning, break-glass)
6. Troubleshooting Guide	Field issues — SSO failures, provisioning drift, de-provisioning, domain/NameID — and remediation
7. Best Practices	Source-of-truth, lifecycle, and pilot-first recommendations

PowerShell scripts

Script	What it does
`GWSSO-Discovery.ps1`	Identity-readiness discovery — inventories the Entra-side gallery app, its SSO mode, SCIM provisioning job, assigned users/groups, and the tenant's verified custom domains, via Microsoft Graph (read-only)

Script	What it does
`GWSSO-Validation.ps1`	Per-user SSO + provisioning validation — for scoped users, checks federated SSO is configured (SAML, NameID = mail), the user is on a verified domain, and is assigned/provisioned; reports pass/exception (read-only)
`GWSSO-Report.ps1`	Identity coverage / exception report — federated vs not, provisioned vs not, on-domain vs `*.onmicrosoft.com`, and the exception list (read-only)
`GWSSO-Remediation.ps1`	Safely fixes scoped Entra-side gaps (assigns a reviewed user/group to the app so provisioning brings them into scope); `-WhatIf` by default
`GWSSO-Set-ScopeTargeting.ps1`	Builds and previews the pilot-scope and owner groups for a staged SSO/provisioning rollout; `-WhatIf` / preview by default

Identity reference doc — outline

The primary architectural deliverable. Recommended structure:

1. **Purpose & scope** — the domains, OUs, and users covered.
2. **Source-of-truth statement** — which directory masters users/groups and the mechanism (SCIM push /

GCDS pull / JIT-convenience).

1. **SSO federation design** — Entra IdP → Google relying party, SAML details, NameID = user.mail,

domain-specific issuer choice.

1. **Domain & identifier map** — verified custom domain(s), UPN ↔ primaryEmail ↔ NameID alignment.
2. **Provisioning design** — attribute mappings, group provisioning, scope (assignment/filter or GCDS

OU rules), and the de-provisioning behavior.

1. **SSO-profile assignment plan** — pilot OU/group first, break-glass = none, expansion batches.
2. **Break-glass** — the Google super-admin outside SSO and the Entra emergency admins.
3. **Exception register** — deliberate deviations (e.g. service accounts not federated), with owner and

review date.

1. **Conformance cadence** — how discovery / validation / report run and how new users/OUs are covered.

What the buyer receives

- This implementation guide and the identity model as a branded Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable process: Assess → Configure → Pilot → Validate → Expand.

Outcome

One identity across Microsoft and Google — users sign in once against Entra ID and reach Google, with users, groups, and licensing kept consistent by a single source of truth and a lifecycle that de-provisions leavers. The identity foundation for coexistence and migration, proven on a named pilot before it touches the whole organization, and reversible at every step.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

GWSSO-Discovery.ps1

Purpose: Read-only discovery of the Microsoft Entra ID side of the Entra->Google identity foundation. It locates the 'Google Cloud / G Suite Connector by Microsoft' enterprise application (service principal), reports its single sign-on mode (SAML / password / none), whether a SCIM provisioning (synchronization) job exists and its state, the count of assigned users and groups, and the tenant's verified custom domains (flagging any onmicrosoft.com reliance). This is the baseline you assess before configuring SSO and provisioning. Google-side state (the SSO profile, provisioned users/groups, GCDS runs) is confirmed in the Google Admin console. HOW TO RUN Install-Module Microsoft.Graph -Scope CurrentUser # once powershell -ExecutionPolicy Bypass -File .\GWSSO-Discovery.ps1 Optional: -AppFilter 'Google' to change the display-name match; -CsvPath .\gwsso-domains.csv. Required Graph scopes (read-only): Application.Read.All, Directory.Read.All, Policy.Read.All, Synchronization.Read.All, User.Read.All. Use 'your-tenant-id' if prompted to select a tenant. EXPECTED RESULTS The Google connector app(s) with SSO mode, provisioning presence/state, assigned user/group counts, and the tenant's verified domains. No changes are made (read-only) — pass. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$AppFilter = 'Google', [string]$CsvPath)

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once powershell -ExecutionPolicy Bypass -File .\GWSSO-Discovery.ps1 Optional: -AppFilter 'Google' to change the display-name match; -CsvPath .\gwsso-domains.csv. Required Graph scopes (read-only): Application.Read.All, Directory.Read.All, Policy.Read.All, Synchronization.Read.All, User.Read.All. Use 'your-tenant-id' if prompted to select a tenant. EXPECTED RESULTS The Google connector app(s) with SSO mode, provisioning presence/state, assigned user/group counts, and the tenant's verified domains. No changes are made (read-only) — pass. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$AppFilter = 'Google', [string]$CsvPath)

GWSSO-Remediation.ps1

Purpose: Applies the safe, reversible remediation for a common Entra-side gap: a reviewed user or group that is NOT assigned to the 'Google Cloud / G Suite Connector by Microsoft' app and therefore is not federated or provisioned. It assigns that principal to the app (the default app role), which brings them into SSO and SCIM provisioning scope on the next cycle. It is intended for a principal you have already reviewed in GWSSO-Report.ps1 and confirmed with the identity owner. Defaults to -WhatIf, so a plain run only PREVIEWS the assignment and changes nothing. Domain moves (onmicrosoft -> custom), NameID mapping, the Google SSO profile, and GCDS are configured in the Entra SSO blade / Google Admin console / GCDS, not by this script. HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\GWSSO-Remediation.ps1 -PrincipalUpn user@your-domain.com powershell -ExecutionPolicy Bypass -File .\GWSSO-Remediation.ps1 -GroupName 'Entra Google SSO Pilot Scope - Pilot' Apply after review: powershell -ExecutionPolicy Bypass -File .\GWSSO-Remediation.ps1 -PrincipalUpn user@your-domain.com -WhatIf:$false Required Graph scopes: AppRoleAssignment.ReadWrite.All, Application.Read.All, Directory.Read.All, User.Read.All, Group.Read.All. Use 'your-tenant-id' if prompted. EXPECTED RESULTS In preview: the exact principal that WOULD be assigned to the Google connector app. When applied: the app-role assignment is created; re-run GWSSO-Validation.ps1 to confirm the user is in scope. Reversible by removing the assignment

(Enterprise app -> Users and groups) in the portal. .NOTES White-label — the repeatable way. Write action — preview (-WhatIf) by default. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([string]$PrincipalUpn, [string]$GroupName, [string]$AppFilter = 'Google')

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\GWSSO-Remediation.ps1 -PrincipalUpn user@your-domain.com powershell -ExecutionPolicy Bypass -File .\GWSSO-Remediation.ps1 -GroupName 'Entra Google SSO Pilot Scope - Pilot' Apply after review: powershell -ExecutionPolicy Bypass -File .\GWSSO-Remediation.ps1 -PrincipalUpn user@your-domain.com -WhatIf:$false Required Graph scopes: AppRoleAssignment.ReadWrite.All, Application.Read.All, Directory.Read.All, User.Read.All, Group.Read.All. Use 'your-tenant-id' if prompted. EXPECTED RESULTS In preview: the exact principal that WOULD be assigned to the Google connector app. When applied: the app-role assignment is created; re-run GWSSO-Validation.ps1 to confirm the user is in scope. Reversible by removing the assignment (Enterprise app -> Users and groups) in the portal. .NOTES White-label — the repeatable way. Write action — preview (-WhatIf) by default. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([string]$PrincipalUpn, [string]$GroupName, [string]$AppFilter = 'Google')

GWSSO-Report.ps1

Purpose: Read-only coverage report for the Entra->Google identity foundation. Across the scoped users it summarizes readiness for federation and provisioning and surfaces the exceptions: users assigned to the Google connector app (in scope for SSO/SCIM) versus not, users on a verified custom domain versus onmicrosoft.com (which Google rejects), and users missing a mail value to serve as the SAML NameID / Google primaryEmail. Use it to build the exception register and to target GWSSO-Remediation.ps1. Google-side provisioned state is confirmed in the Google Admin console. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWSSO-Report.ps1 Optional: -AppFilter 'Google'; -CsvPath .\gwsso-coverage.csv to export the per-user coverage. Required Graph scopes (read-only): Application.Read.All, Directory.Read.All, Policy.Read.All, Synchronization.Read.All, User.Read.All, Group.Read.All. Use 'your-tenant-id' if prompted. EXPECTED RESULTS A coverage summary (in scope / on custom domain / mail present) and exception lists (not assigned, onmicrosoft UPN, missing mail). No changes are made — pass. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$AppFilter = 'Google', [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\GWSSO-Report.ps1 Optional: -AppFilter 'Google'; -CsvPath .\gwsso-coverage.csv to export the per-user coverage. Required Graph scopes (read-only): Application.Read.All, Directory.Read.All, Policy.Read.All, Synchronization.Read.All, User.Read.All, Group.Read.All. Use 'your-tenant-id' if prompted. EXPECTED RESULTS A coverage summary (in scope / on custom domain / mail present) and exception lists (not assigned, onmicrosoft UPN, missing mail). No changes are made — pass. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$AppFilter = 'Google', [string]$CsvPath)

GWSSO-Set-ScopeTargeting.ps1

Purpose: Creates (or previews) the security groups this solution targets for a staged SSO + provisioning rollout: the pilot scope 'Entra Google SSO Pilot Scope - Pilot' (the representative users you validate federation and provisioning on before org-wide enforcement) and the owner group 'Entra Google SSO Owners - Tenant - Steady' (the identity admins who own the source-of-truth decision and review exceptions). Assign the pilot-scope group to the Google connector app so only its members federate and provision until the gate passes; mirror it to the matching Google OU/group and scope the Google SSO

profile the same way. By default it PREVIEWS (no changes); pass -Create to create groups that do not yet exist. HOW TO RUN Preview which groups exist / would be created (read-only): powershell -ExecutionPolicy Bypass -File .\GWSSO-Set-ScopeTargeting.ps1 Create the groups if missing: powershell -ExecutionPolicy Bypass -File .\GWSSO-Set-ScopeTargeting.ps1 -Create Required Graph scopes: Group.Read.All + Directory.Read.All (preview); Group.ReadWrite.All + Directory.Read.All (with -Create). Use 'your-tenant-id' if prompted. EXPECTED RESULTS In preview: whether each named group exists and its member count. With -Create: the missing groups are created (empty) and their object IDs returned so you can add members. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph unless -Create is supplied. #> [CmdletBinding(SupportsShouldProcess = $true)] param([switch]$Create)

How to run: Preview which groups exist / would be created (read-only): powershell -ExecutionPolicy Bypass -File .\GWSSO-Set-ScopeTargeting.ps1 Create the groups if missing: powershell -ExecutionPolicy Bypass -File .\GWSSO-Set-ScopeTargeting.ps1 -Create Required Graph scopes: Group.Read.All + Directory.Read.All (preview); Group.ReadWrite.All + Directory.Read.All (with -Create). Use 'your-tenant-id' if prompted. EXPECTED RESULTS In preview: whether each named group exists and its member count. With -Create: the missing groups are created (empty) and their object IDs returned so you can add members. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph unless -Create is supplied. #> [CmdletBinding(SupportsShouldProcess = $true)] param([switch]$Create)

GWSSO-Validation.ps1

Purpose: Read-only, per-user conformance check for Entra->Google SSO and provisioning. For the scoped users (a pilot group, or the whole tenant) it validates the identity-readiness signals: the user is on a verified custom domain (not onmicrosoft.com), has a usable mail value to serve as the SAML NameID / Google primaryEmail, and is assigned to the Google connector app (so SSO and SCIM provisioning bring them into scope). It also confirms the app-level standards once: SSO mode is SAML and a provisioning job exists and is running. Each user is reported CONFORMS or with the specific issue. Google-side sign-in and provisioned state are confirmed in the Google Admin console; this validates the Entra plane. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWSSO-Validation.ps1 Scope to a pilot: -PilotGroup 'Entra Google SSO Pilot Scope - Pilot' Change the connector match: -AppFilter 'Google' Required Graph scopes (read-only): Application.Read.All, Directory.Read.All, Policy.Read.All, Synchronization.Read.All, User.Read.All, Group.Read.All. Use 'your-tenant-id' if prompted. EXPECTED RESULTS An app-level summary (SSO mode SAML, provisioning running) plus a per-user pass/fail: CONFORMS, or issues (onmicrosoft domain, missing mail/NameID source, not assigned to the app). No changes. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$AppFilter = 'Google', [string]$PilotGroup)

How to run: powershell -ExecutionPolicy Bypass -File .\GWSSO-Validation.ps1 Scope to a pilot: -PilotGroup 'Entra Google SSO Pilot Scope - Pilot' Change the connector match: -AppFilter 'Google' Required Graph scopes (read-only): Application.Read.All, Directory.Read.All, Policy.Read.All, Synchronization.Read.All, User.Read.All, Group.Read.All. Use 'your-tenant-id' if prompted. EXPECTED RESULTS An app-level summary (SSO mode SAML, provisioning running) plus a per-user pass/fail: CONFORMS, or issues (onmicrosoft domain, missing mail/NameID source, not assigned to the app). No changes. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$AppFilter = 'Google', [string]$PilotGroup)